Engine Model(s): PT6A

74-10-01

IGNITION CURRENT REGULATOR - DESCRIPTION AND OPERATION

1. Description and Operation

The ignition current regulator, mounted on the left-hand side of the accessory gearbox, is a box like unit comprising two sets of ballast tubes encased in protective cushions, which are mounted on the inner face of the regulator cover. Compression springs mounted in the box of the regulator retain the tubes in their respective receptacles.

The electrical circuitry is designed to provide a selection of either set of ballast tubes, interconnected to the respective glow plug. The selected set of ballast tubes provide an initial current surge to the glow plug when switched on, which stabilizes to a constant value for the tubes in approximately 30 seconds. This characteristic provides a rapidly heated glow plug for fast light-ups.

The ballast tubes contain helium and hydrogen gases, and a pure iron filament. The filament, having a positive coefficient of resistance, provides a stabilizing effect on the current passing through it, thereby controlling the current to a nearly constant value over a wide range of voltages.

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